

Xingjian Zhang

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Education

University of Michigan, School of Information, *Advised by Prof. Qiaozhu Mei* Sep 2022 – Present
Ph.D. in Information Science **GPA: 3.89/4.00** Ann Arbor, MI

University of Michigan, EECS Department Sep 2020 – Apr 2022
B.S. in Computer Science; Minor in Mathematics **GPA: 3.90/4.00** Ann Arbor, MI

Work Experience

Research Scientist Intern - Meta Platforms, Inc., Menlo Park, CA May 2026 – Oct 2026

- **Monte Carlo Guided Tree Search for LLM Recursive Improvement**
 - Developed an efficient guided-sampling pipeline for synthetic on-policy data generation.
 - Contributed personal productivity domain data for Muse post-training, Meta's LLM.

Research Intern - Google DeepMind & YouTube Shorts, Mountain View, CA May 2025 – Aug 2025

- **Multimodal Tool-Using Agent for Video Event Retrieval**
 - Proposed an end-to-end Reinforcement Learning framework to train a multimodal agent to use tools.
 - Built benchmark data and core multimodal/tool-use components in the open-source project `simply`. [4]

Research Intern - Google DeepMind & YouTube, Mountain View, CA May 2023 – Aug 2023

- **Efficient Transformer Fine-tuning and Serving** [13]
 - Proposed a GPU/TPU-friendly adaptive layer-skipping algorithm based on input sequence complexity.
 - Achieved the same performance with 60% compute and delivered the algorithm in TensorFlow Recommenders (TFRS).

Graduate Student Research Assistant - University of Michigan School of Information Sep 2022 – Present

- **Foundation Model for Scientific Innovation** [7–10, 12, 14]
 - Conduct research on LLM reasoning, evaluation, and scientific workflow automation; design benchmarks, curate scientific-literature datasets, implement reinforcement learning and agent systems, and run computational experiments. [8, 9, 14]
- **Graph Machine Learning** [3, 5, 6, 11]
 - Built metadata-rich graph learning benchmarks and developed methods/analyses for partial-ranking likelihood estimation, causal inference, and LLM use of graph structure. [3, 5, 6]

Open Source Projects

- [google-deepmind/simply](#): JAX codebase for rapid iteration on LLM research.
- [TRAIS-Lab/dattri](#): PyTorch library for efficient data attribution.
- [Graph-Learning-Benchmarks/gli](#): Metadata-rich graph learning benchmark platform.

Selected Awards

Rackham Doctoral Fellowship	Sep 2022	James B. Angell Scholar	Mar 2022
Tang Junyuan Scholarship	Oct 2021	J. Wu & J. Sun Sunshine Scholarship	Oct 2020

Technical Skills

- **Programming Language:** Python (Professional); C, C++, R, Matlab, Mathematica.
- **Framework and Tools:** JAX, PyTorch, TF, Deep Graph Library (DGL), TensorFlow Recommenders.

Additional Experience

- **Workshop Organizer:** Graph Learning Benchmark@KDD'23, LLMs for Individuals, Groups, and Society@WSDM'24.
- **Graduate Student Instructor (GSI):** Data Mining, Building Interactive Applications, Introduction to Information Analysis.

Publications/Preprints

- [1] Junwei Deng, Ting-Wei Li, Shiyuan Zhang, Shixuan Liu, Yijun Pan, Hao Huang, Xinhe Wang, Pingbang Hu, Xingjian Zhang, and Jiaqi W Ma. 2024. dattri: A Library for Efficient Data Attribution. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track (NeurIPS)*, 2024. Retrieved from <https://neurips.cc/virtual/2024/poster/97763>
- [2] Benhao Huang, Yingzhuo Yu, Jin Huang, Xingjian Zhang, and Jiaqi Ma. 2025. DCA-Bench: A Benchmark for Dataset Curation Agents. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining Datasets and Benchmarks Track (KDD)*, 2025. Retrieved from <https://arxiv.org/abs/2406.07275>
- [3] Jin Huang, Xingjian Zhang, Qiaozhu Mei, and Jiaqi Ma. 2024. Can LLMs effectively leverage graph structural information: when and why. *Transactions on Machine Learning Research (TMLR)* (2024). Retrieved from <https://arxiv.org/pdf/2309.16595.pdf>
- [4] Chen Liang, Da Huang, Chengrun Yang, Xiaomeng Yang, Andrew Li, Xinchun Yan, Alex Zhai, Xingjian Zhang, Jiayi Tang, Lizhang Chen, and Ran Tian. 2025. Simply: an experiment to accelerate and automate AI research. Retrieved from <https://github.com/google-deepmind/simply>
- [5] Jiaqi Ma*, Xingjian Zhang*, and Qiaozhu Mei. 2022. Fast learning of MNL model from general partial rankings with application to network formation modeling. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM)*, 2022. 715–725. Retrieved from <https://dl.acm.org/doi/abs/10.1145/3488560.3498506>
- [6] Jiaqi Ma*, Xingjian Zhang*, Hezheng Fan, Jin Huang, Tianyue Li, Ting Wei Li, Yiwen Tu, Chenshu Zhu, and Qiaozhu Mei. 2022. Graph learning indexer: A contributor-friendly and metadata-rich platform for graph learning benchmarks. In *Proceedings of the Learning on Graphs Conference (LoG)*, 2022. 7–1. Retrieved from <https://arxiv.org/pdf/2112.15575.pdf>
- [7] Xiyuan Wang, Jin Huang, Xingjian Zhang, Tianhao Wang, and Jiaqi W Ma. 2026. Your Reasoning Benchmark May Not Test Reasoning: Revealing Perception Bottleneck in Abstract Reasoning Benchmarks. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. Retrieved from <https://arxiv.org/abs/2512.21329>
- [8] Yutong Xie, Zhuoheng Li, Xiyuan Wang, Yijun Pan, Qijia Liu, Xingzhi Cui, Kuang-Yu Lo, Ruoyi Gao, Xingjian Zhang, Jin Huang, Walter Yuan, Matthew O Jackson, and Qiaozhu Mei. 2025. Be.FM: Open foundation models for human behavior. *arXiv [cs.AI]* (May 2025).
- [9] Xingjian Zhang*, Yutong Xie*, Jin Huang, Jing Ma, Zhaoying Pan, Qijia Liu, Ziyang Xiong, Tolga Ergen, Dongsub Shim, Honglak Lee, and others. 2025. MASSW: A New Dataset and Benchmark Tasks for AI-Assisted Scientific Workflows. In *Findings of the Association for Computational Linguistics: NAACL*, 2025. Retrieved from <https://arxiv.org/abs/2406.06357>
- [10] Xingjian Zhang, Tianhong Gao, Suliang Jin, Tianhao Wang, Teng Ye, Eytan Adar, and Qiaozhu Mei. 2026. Through the Judge's Eyes: Inferred Thinking Traces Improve Reliability of LLM Raters. *Transactions on Machine Learning Research* (2026). Retrieved from <https://openreview.net/forum?id=1jLQ629Yps>
- [11] Xingjian Zhang, Shixuan Liu, Yixin Wang, and Qiaozhu Mei. 2024. Leveraging LLM-Generated Structural Prior for Causal Inference with Concurrent Causes. In *Proceedings of the Causality and Large Models Workshop at NeurIPS 2024 (NeurIPS Workshop)*, 2024. Retrieved from <https://openreview.net/forum?id=AqRQyOINf8>
- [12] Xingjian Zhang, Sophia Moylan, Ziyang Xiong, Qiaozhu Mei, Yichen Luo, and Jiaqi W Ma. 2026. FlyAOC: Evaluating Agentic Ontology Curation of Drosophila Scientific Knowledge Bases. *arXiv preprint arXiv:2602.09163* (2026).
- [13] Xingjian Zhang, Jiayi Tang, Yang Liu, Xinyang Yi, Li Wei, Lichan Hong, Qiaozhu Mei, and Ed H Chi. 2024. Conditional Transformer Fine-Tuning by Adaptive Layer Skipping. In *Proceedings of the 5th Workshop on Practical ML for Limited/Low Resource Settings (ML4LowResource)*, 2024. Retrieved from <https://openreview.net/forum?id=Rd0lJC3c9a>
- [14] Xingjian Zhang, Ziyang Xiong, Shixuan Liu, Yutong Xie, Tolga Ergen, Dongsub Shim, Hua Xu, Honglak Lee, and Qiaozhu Mei. 2025. MapExplorer: New Content Generation from Low-Dimensional Visualizations. In *Proceedings*

of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025. Retrieved from <https://arxiv.org/abs/2412.18673>